

Web Application Penetration Testing Report

Executive Summary

A web application security assessment was performed against DVWA hosted on the Metasploitable2 environment. The objective was to identify common OWASP Top 10 vulnerabilities and evaluate the application's security posture.

Target

DVWA

URL: <http://192.168.56.101/dvwa>

Findings

Test ID: 001

Vulnerability:
SQL Injection

Severity:
Critical

Description:

User-supplied input was processed without proper validation, allowing database queries to be manipulated.

Impact:

An attacker could retrieve sensitive information, bypass authentication, and potentially compromise the database.

Recommendation:

Use parameterized queries and input validation.

Test ID: 002

Vulnerability:
Reflected Cross-Site Scripting (XSS)

Severity:
Medium

Description:
The application reflected unsanitized user input back to the browser.

Impact:
An attacker could execute malicious JavaScript within a victim's browser session.

Recommendation:
Implement output encoding and input sanitization.

Burp Suite Assessment

Burp Suite Community Edition was used to intercept and analyze HTTP requests between the browser and the application.

Observations:

- Login requests were inspected.
- Request parameters were analyzed.
- Application traffic was reviewed.

OWASP ZAP Assessment

OWASP ZAP was used to perform automated scanning and identify potential weaknesses within the application.

Findings Table

Finding ID	Vulnerability	CVSS Score	Remediation
F001	SQL Injection	9.1	Parameterized Queries
F002	Reflected XSS	6.1	Input Validation

Web Test Summary

The assessment identified SQL Injection and Cross-Site Scripting vulnerabilities within the test application. These weaknesses could allow attackers to access sensitive information and execute malicious scripts. Proper input validation, output encoding, and secure development practices are recommended.